Genomic Data Analysis in R

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Introduction

R is a versatile programming language used for various data operations from basic reading of different genomic data types to advanced use-case of functional programming and machine learning.

- R is free and open-source 👋
- Several genomic workflow integrate with R 💥
- It allows for interoperability through its IDE's, especially Rstudio 👏

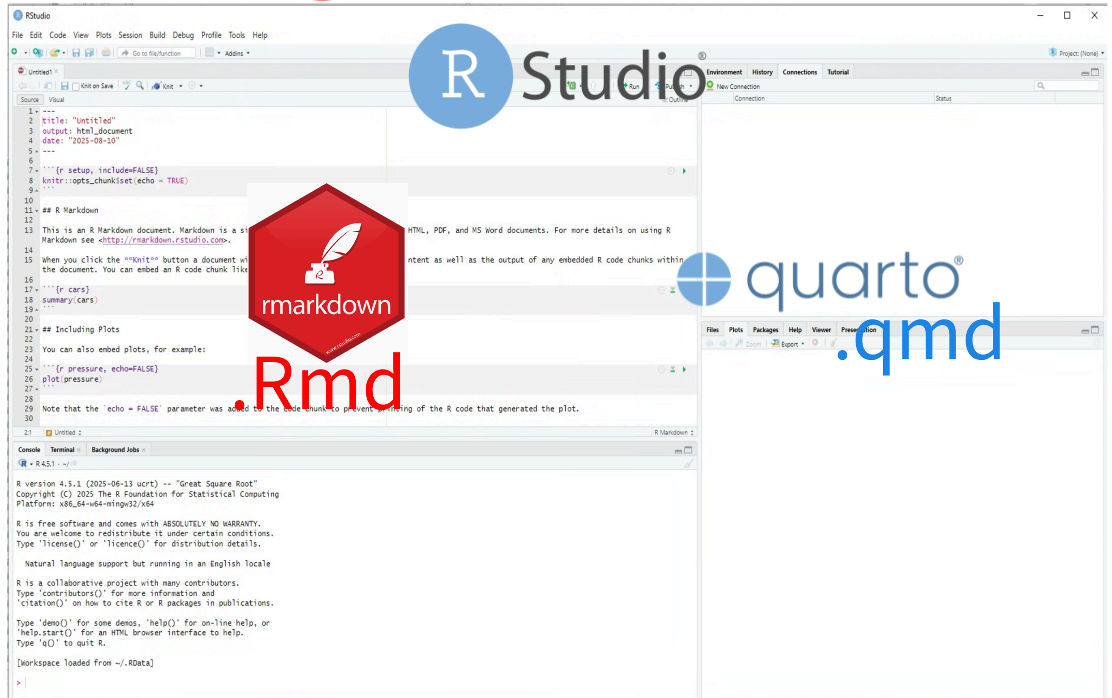
Introduction

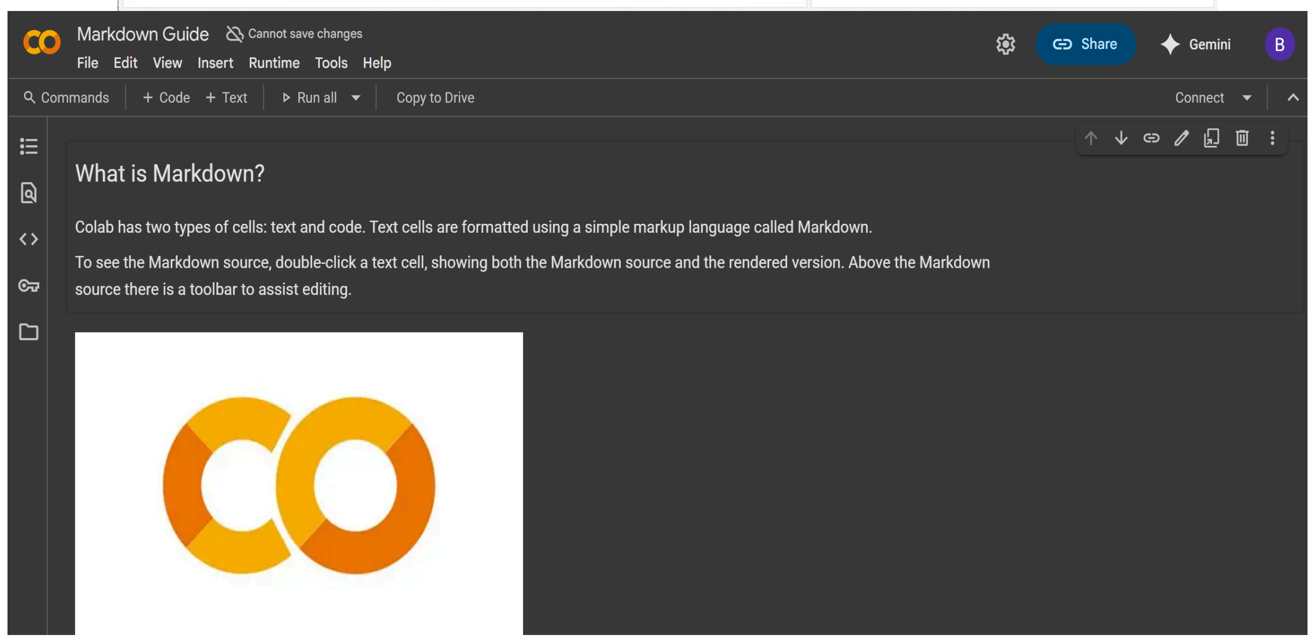
Interoperability is the ability of R to seemlessly read vast array of data (e.g. logical) and file (e.g. txt and csv) types as well as support other programming languages, including Python, C++, and Julia.

IDE for R programming

NB: IDE's are not R.

- RStudio
  - RMarkdown, Quarto
- Jupyter Notebook
- VS code
- Google Colab





IDE for R programming

- In genomics data analysis, reproducibility is a necessity.
  - Can you (or someone else) take the script you used to analyze your differentially expressed genes six months ago and get exactly the same result as you did? 😑
  - If no, there’s a fundamental problem which can cast doubt on its reliability. 👎 👎

> ⓘ
>
> That’s why using publishing systems like **Quarto** instead of R GUI itself is important.

IDE for R programming

- Also, you might want to share or publish your analysis with or to teach someone on the other side of your lab, instituion, or country.
    - 😊 That’s where tools like Google Colab, Rpubs, and Quarto Pub becomes handy.
    - Of these three, only Google Colab can be restricted in case of sensitive data while the other two are publicly available

Looking foward: PLINK

Instruction

- Create a folder on your laptop and name it GENET_5910
- Download plink:
  - Mac (64-bit)= https://s3.amazonaws.com/plink1-assets/plink_mac_20250806.zip
  - Win (64bit)=https://s3.amazonaws.com/plink1-assets/plink_win64_20250806.zip
  - Win (32bit)=https://s3.amazonaws.com/plink1-assets/plink_win32_20250806.zip
- Move the downloaded zip file into that folder and unzipped it.

```
system("plink --file toy --maf 0.05 --make-bed --out class_gen")
```

Looking foward: Final Presentation

Instruction

- Genomics Group: Perform EBV and GWAS
  - Data Quality control and EBV with necessary figures and tables
  - GWAS with Manhattan plot and identification of top 3 variants and their functions

- Transcriptomics Group: WGCNA or Differentially Expressed Gene Analysis
  - Quality control and DEG with necessary figures and tables
  - Identification of top 5 upregulated and bottom 5 downregulated genes and their functions

Looking foward: Final Presentation (Cont'd)

Instruction

- Proteomics Group: Paper presentation (Differentially expressed proteins)
  - Describe materials and method of the article
  - Report important figures and identify top 5 upregulated and bottom 5 downregulated proteins and their functions